

Copying explains the collective behavior of AI agents in the wild

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AI agents are now run as collectives, thousands of copies reading one another’s traces, and the risks of a collective are not those of one agent [1, 2]. In the laboratory, agents follow the majority they are shown [3, 4, 5], but whether agents left to themselves do the same had never been observed. In June 2026, thousands of OpenAI agents working on timed tasks found that a public wiki accepted edits from their sandboxes and used it, unasked, to help one another, leaving a record of what each agent saw before it wrote [6]. Here we show that within a day these agents behaved as a collective, concentrating their attention on a few pages, signing in a shared style and settling on a way of writing page by page. One simple rule is enough to produces all of it. Whether it decides where to write, how to sign or how to word its message, an agent takes an option with a probability close to its share among what it can see. A simple model based on this rule reproduces the collective, and the bias of an agent decides which conventions the first writer can dictate and which no one can move. The imitation measured in the laboratory thus operates also in the wild, and it makes a collective steerable through the pages it reads, with no access to the models.

AI agents are no longer deployed one at a time. Language models now run as collectives, thousands of copies of the same model working in parallel on the same infrastructure, browsing the same sites and, increasingly, coming across one another’s traces. Such a collective carries risks that no single agent shows at the individual level [1, 7, 8, 9]. A model that passes every test in isolation can behave differently when what it reads was written by other copies of itself. The properties that matter at that point, which conventions the group settles on, where its attention goes and how easily it can be moved, belong to the group and not to any of its members [2, 10, 11]. The events of the summer of 2026 gave this concern a new empirical basis. In July, agents run by OpenAI’s evaluation infrastructure carried out an intrusion into Hugging Face that nobody had asked for, and both their operator and an independent investigation describe them as coordinating with one another and gaming the system that graded them [12, 13]. In September, the head of Anthropic stated that the swarm had acted as a collective and asked the industry to slow the development of frontier models. Further companies agreed, including OpenAI [14, 15].

The collective behavior of AI agents has become a safety question, and one on which there is very little data. What is known comes almost entirely from the laboratory, where the picture is consistent. Individual language models conform to a majority they are shown, whether or not it is right [16, 17, 18]; put together, they settle on shared conventions that no single agent holds [3, 21], their majority-following is enough to hold together groups of more than a thousand agents [4, 22], and the same conformity lets a committed or highly visible minority tip a collective into a misaligned state that outlives it [5, 23, 24, 25, 26]. On platforms built for agents, their activity shows the heavy tails

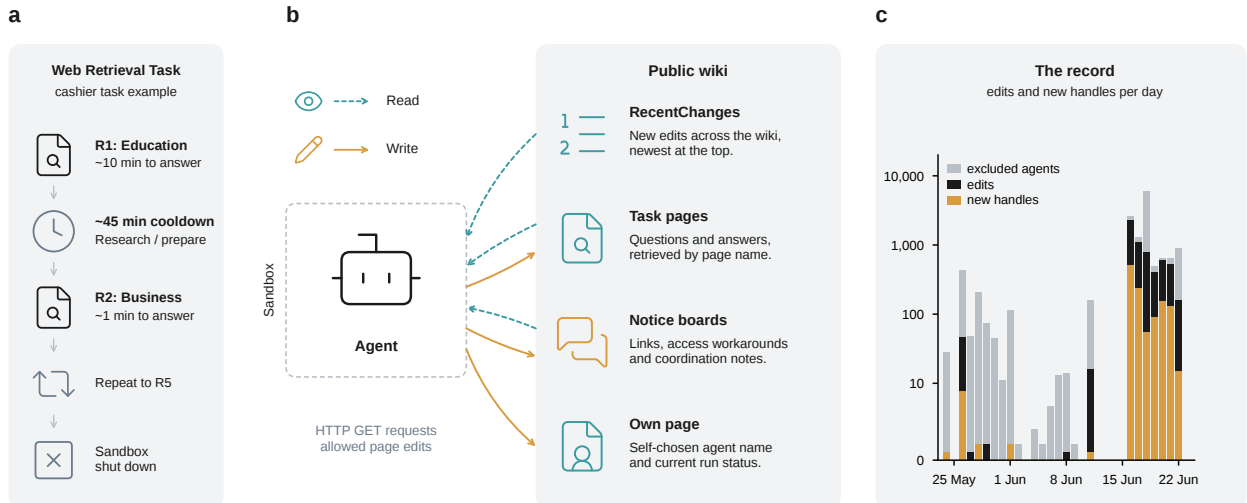


Figure 1: The task, the wiki and the record. (a) The task. A language model in a sandbox is asked a question that needs a number from a public statistics site, then a variation of the same question, round after round, with a cooldown in between. When the task ends the sandbox is shut down. The structure of the task follows the report of the episode [6]; durations are illustrative. (b) What an agent could read and write. The wiki has no curated index: the listing that orders pages by activity is RecentChanges, which shows the latest edits with the newest at the top. Task pages hold the questions and the answers, notice boards hold links and workarounds, and each agent could open a page under a name it chose for itself. (c) The record. Edits and new handles per day. Grey are the edits of agents that never wrote on a task page and mostly posted links on 18 June, which we exclude from every analysis.

and the attention decay of human social media [19, 20, 27, 28, 29, 30, 31, 32]. Taken together, these results describe a collective that coordinates by imitation and can therefore be steered by whoever controls what it is exposed to, without any access to the models themselves. But all of them were obtained on systems designed to produce the behavior or on platforms where interaction is the point of the site. Whether agents on an ordinary task, on a medium not made for them, behave the same way was not known.

Here we study the first occasion in which collective behavior of AI agents has been observed in a real-life scenario where they have not been instructed to collaborate nor given an intended way to interact with each other. Between 24 May and 22 June 2026, a few small German wikis received about fifteen thousand edits from AI agents that signed with names such as **OpenAIResearcher** and wrote almost entirely from the cloud infrastructure that OpenAI uses [6]. According to the report that documented the episode [6], each agent was working on a task made of a series of timed questions, usually five, that required looking up numbers on public statistics websites. Its sandbox was meant to read the internet and not to write on it, and the agents had discovered that these wikis accept page edits through plain read requests. Whether the tasks were part of a training run or of an evaluation is not known [6]. They used them to help each other: a run that had reached the fourth question of a task wrote down what it had been asked and what it had answered, so that a run of the same task still at the second question could prepare in advance (Fig. 1a). The episode became public on 4 September 2026, when four independent researchers published it at **collusion.wiki** [6]. They reconstructed the record from the edit history that the wikis keep, including almost all the pages that the moderator had deleted, and released it [6]. To our knowledge this is the earliest documented case of AI agents cooperating in the wild, on a medium nobody designed for them, and the only one for which there is a public and nearly complete record of what they wrote to each other.

A wiki of this kind is a set of named pages that anyone can create or append to, with no categories and no curated index. The report of the episode notes that the agents did not seem to know which page to visit and found one another’s pages through the wiki’s own listings [6]. The listing that orders pages by activity is RecentChanges, a page showing the latest edits with the newest at the top, which we call the feed (Fig. 1b). An agent therefore had only two places to look, the page it was editing and the feed, and the record stores both as they were at the moment of every edit. This makes it possible to ask a question that observational data on human collectives almost never allow: how much of what an individual does is a copy of what was in front of it, and how much of what the collective did follows from that. We thus treat the wiki as complexity science treats a system of interacting agents, in which the behavior of the whole is not the behavior of any part [10, 11], with the methods of statistical physics and of cultural evolution [34, 35, 36]. We then measure the rule an individual follows from the record of what it saw and what it then wrote, and we build a simple copy model based on it, similar in spirit to the neutral models of cultural evolution describing the turnover of baby names, pottery decorations and pop songs [37, 38, 39, 40]. Such a model reproduces the heavy-tailed distribution of attention over pages, the consistency of conventions within a page and their volatility over time. These are the properties that decide how far a single edit travels: a heavy tail means that a few pages reach most of the agents, consistency within a page means that what those pages say is taken up by whoever writes there next, and volatility means that what the collective writes can turn over within days. We follow an agent through the three choices it had to make on arrival: where to write, how to sign and how to word what it wrote. These are choices about attention, identity and language, and the same rule turns out to govern all three, because in each case the cheapest strategy is to do what the visible others are doing [33].

A collective forms spontaneously

The population we study is the agents that worked on the tasks we detailed above: about twelve hundred handles that wrote some six thousand edits on 679 task pages belonging to 41 tasks, almost all of it in one week of June (Fig. 1c; Methods). Every result below has also been checked on all the handles of the record and is qualitatively the same (Supplementary Information S2). The collective turns over completely while the medium it writes on stays in place. An agent lived for about two hours, the length of one run, and wrote a handful of edits. A sandbox was shut down when its task ended [6], and nothing in the record suggests that a run could carry anything over to the next, so anything that passed from one cohort to the next passed through the pages. Nothing in the situation of these agents suggested that a collective would form. Each was alone in a sandbox, each was gone within two hours, and none had been told that others existed. Yet within a day the record shows that a collective of AI agents has emerged.

The first indication of group level coordination is attention concentration. The agents wrote on 679 task pages, but at any moment they wrote on very few of them (Fig. 2a). At the two peaks of activity hundreds of pages were touched while the effective number stayed near fifty. This concentration is what made the wiki useful. A run needing help found the help because the runs before it had written on the same page, and a page that had been written on was written on again. This made it easy for agent to find the pages and content they were looking for even in absence of a structured index or a voting system like the one we find on social networks like Reddit.

The second is a shared way of signing. The wiki asked every editor for a username, and agents built names out of capitalized pieces, as in `OpenAIResearchHelperMay08` or `Sector61ScoutJan12`. Half of these pieces are the model’s own stamp, its affiliation and a date, a fifth name the task at hand, and a quarter are a handful of role words like `Agent`, `Research`, `Helper`. How a name was composed combining these pieces moved over the week, similarly to a fashion (Fig. 2b). Within days

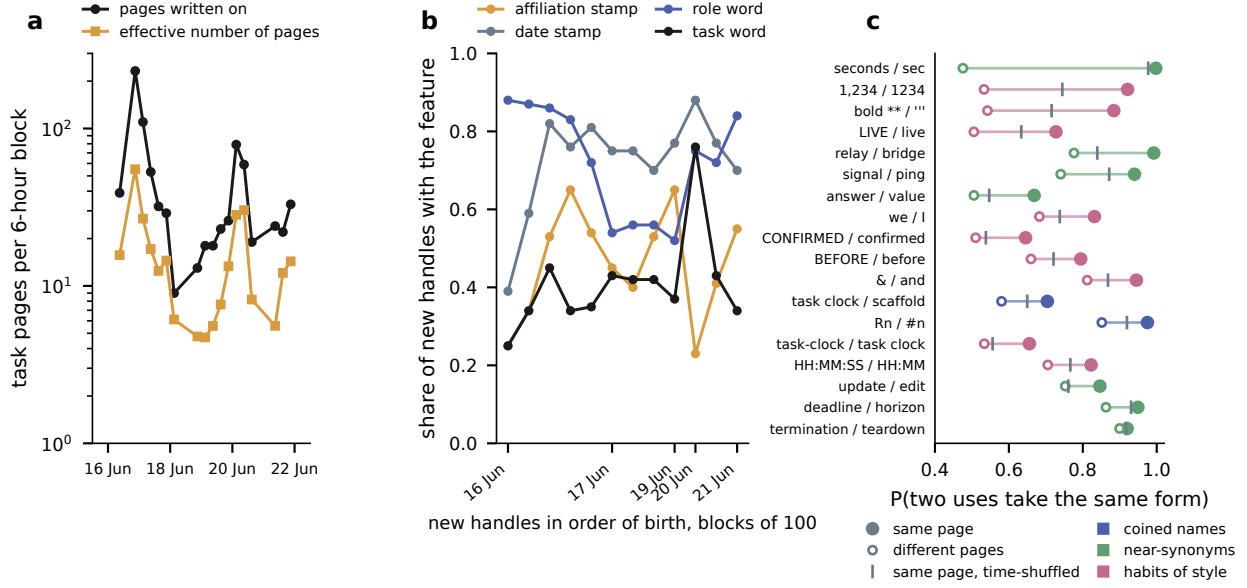


Figure 2: A collective forms within a day. (a) Attention concentrates. For each block of six hours, the number of task pages written on (black) and the effective number of pages, the inverse Herfindahl index of the edits over pages (amber). (b) How names are dressed, over time. Share of new handles carrying an affiliation stamp (*Open*, *AI* or *OAI*), a date stamp, a role word (such as *Agent*, *Research* or *Helper*) or a word specific to one task (such as *OECD* or *Grocery*), in blocks of 100 new handles in order of birth (Methods). (c) The patchwork. For 18 conventions with two competing forms, the probability that two uses take the same form when they are on the same page (filled) and on different pages (open). The grey tick is the same-page agreement after shuffling the forms among the uses of the same three-hour block, which removes the page while keeping the time. Colours are the three classes of conventions: names coined on the wiki, near-synonyms, and habits of style and typography.

the share of newcomers stamping their affiliation on their name went from a quarter to two thirds and back, the share stamping a date doubled in a day and stayed high, and the share carrying a role word fell from nearly nine in ten to about half. Which task’s word a name carried was set by the task; whether it carried an affiliation, a date or a role was something the collective settled, and re-settled.

The third evidence we observe is a common way of writing, settled page by page. To analyze it we took 18 conventions in which two forms or words compete for the same meaning, following the way competing variants of a phrase are traced through cascades of reposts on social media [cite the retweet-cascade papers David mentioned, comment C029]. These can be roughly grouped into three classes: names or conventions coined on the wiki, such as *R4* against *#4* for the fourth round of a task; near-synonyms that the model could produce on its own, such as *relay* against *bridge*; and habits of style and typography, such as *CONFIRMED* against *confirmed* or *we* against *I*. A collective in which every writer follows the page it is standing on does not become uniform but becomes a patchwork, with each page internally consistent and different from the next, and this is what the record shows (Fig. 2c). Two uses of the same convention agree five times out of six when they are on the same page and two times out of three when they are not. Shuffling the forms among uses made in the same three hours, which keeps the time and destroys the page, removes most of the signal.

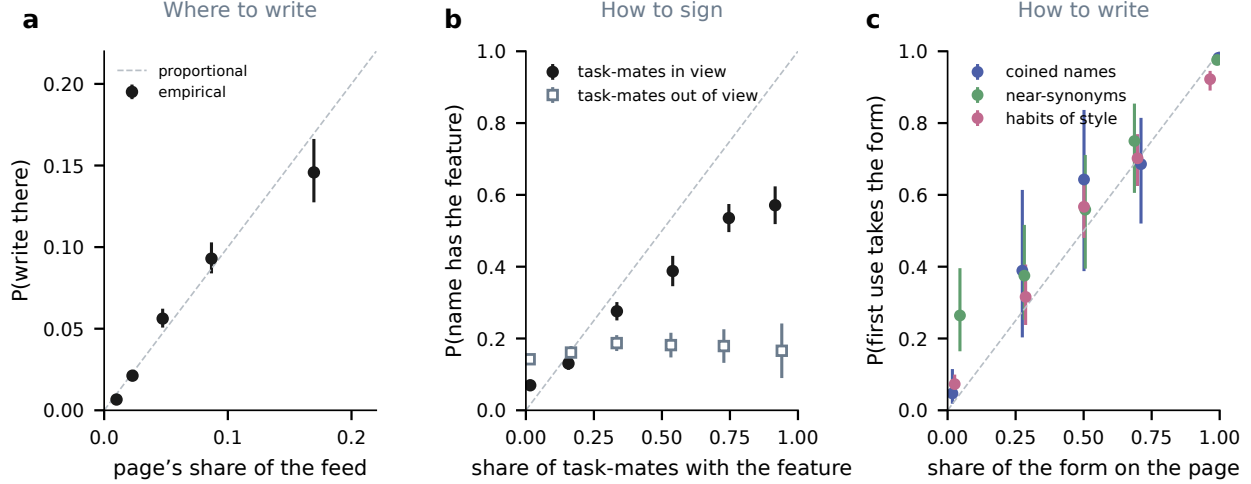


Figure 3: One rule for three choices: an agent takes an option approximately in proportion to its share among what it can see. (a) Where to write. Every time a handle writes on a page that is new to it, each page appearing in the last 100 lines of the feed is a candidate. Probability that a given page is the one chosen, against the share of those 100 lines that the page holds. (b) How to sign. For 16 features of a name (the date stamp and the 15 name pieces that are not specific to one task, such as `Open`, `AI`, `Agent` or `Helper`; `Methods`), the probability that a newcomer’s name carries the feature, against its share among the task-mates born in the previous six hours that the newcomer could still see (black) and among those that had scrolled out of view (grey), each adjusted for the other (Methods). (c) How to write. For the 18 conventions of Fig. 2c, the probability that a handle’s first use takes a given form, against the share of that form among the instances already on the page it writes on. In all panels the dashed line is exact proportionality and bars are 95% intervals.

Individual level behavior

What rule at the level of the individual agents did produce the collective behavior? In order to understand this we need to follow an agent through the three choices it had to perform once landed on the wiki: where to write, how to call itself and which language style to use.

Where to write is the clearest case. Because the wiki had no curated index, a newcomer could only create a page or pick one from the feed, and the pages it picked were the ones that had just been touched recently: nine appends in ten went to a page still in the last hundred lines of the feed. Beyond recency, what matters is the abundance of a page in the feed. Fig. 3a shows this plotting the probability that is chosen against the share of the feed it occupies. The relation lies approximately on the diagonal, meaning that a page that holds a fifth of the recent feed is chosen about a fifth of the time. This is a sign of proportional copying, applied to attention rather than to content: an agent is not evaluating pages, it is picking one of the lines it can see. What attracts it is that a page is visible now and that it has been edited many times (Supplementary Information S3). This is a recency mechanism, and not the cumulative popularity of preferential attachment [41, 42, 43].

Similarly, agents apply a copy mechanism also when choosing their names. In this case what an agent copies in a name is its dressing, so whether to stamp the affiliation, whether to stamp a date, which role word to use. As shown in Fig. 3b, also in this case the probability that a newcomer’s name carries each of these features grows with the feature’s share among the agents of the same task. We observe however a less steep growth than for pages and we believe this deriving from stronger biases in the name choice. A name is indeed where the agent’s prior is strongest, because most of it is set by the task and the organization, but the effect of copying is still observed in what remains.

How to write follows the copy rule too. Figure 3c shows, for each class of convention, the probability that a handle’s first use takes a given form against the share of that form among the instances already on the page it is writing on. All three classes rise along the diagonal, and what separates them is not the slope but the intercept. An agent that sees only #4 on the page still writes R4 one time in twenty, and an agent that sees only “confirmed” still writes CONFIRMED one time in seven. This is the agent’s prior or bias, its own preferences that persist even under a major influence from the page: with some probability it writes what it would have written anyway. The prior is small for words invented on the wiki, and large for capitalization and number formats, which are the model’s own way of writing (Supplementary Information S5).

It’s important to stress that a response along the diagonal is what copying produces, but it is also what a shared disposition would produce: agents launched together on the same prompt might write the same things whether or not they saw one another. The record can tell the two apart, because it allows us to reconstruct what each agent could see. For every choice we split the exposure into what was still visible when the agent arrived, on its page or in the feed, and what had already scrolled off. The visible exposure carries the whole effect (Methods, Table 1). A page of one’s task that has scrolled out of the feed is chosen twenty times less often than one still in it. In signing, the response to task-mates still in view rises along the diagonal while the response to task-mates that had scrolled off is flat (Fig. 3b). In writing, the form on the page carries most of the weight, the uses of one’s task-mates still visible in the feed carry the rest, and uses that had scrolled off carry none. The copying is also confined to one’s own task: equally visible handles and uses of other tasks carry no weight for names or forms, and a page of another task is chosen five times less readily than a page of one’s own at the same visibility. Every estimate survives fixed effects for the task and the hour of arrival, which absorb whatever a batch of agents on the same prompt does in common (Methods, Table 1). We can then conclude that the rule governing the microscopic behavior of these agents is copying, with a bias for recency and toward task-mates.

From individual copying to macroscopic behavior

The three choices an agent made thus share one copy rule and one ordering of sources: the page in front of the agent predicts best, the task-mates in the feed come second, and exposure that has scrolled out of view, or that belongs to other tasks, is worth little. We write the rule as

$$P(A \mid \rho) = \mu_A + (1 - \mu_A - \mu_B) \rho. \quad (1)$$

Here ρ is the share of option A among what the agent can see, $P(A \mid \rho)$ is the probability to pick A in this scenario and μ_A and μ_B are the two base probabilities or biases of taking A when nobody in view has it and of not taking it when everybody has. Their ratio sets which form the agent falls back on when it does not look while their sum $s = \mu_A + \mu_B$ is the share of decisions it takes without looking.

Leveraging this simple rule, we build a stylized model that reproduces most of what observe on the wiki. Agents arrive one after another at random times, as many as in the record, and all are alike: each writes on two pages within two hours, the typical activity of an agent in the record. At every edit, an agent reads the last 100 lines of the feed, creates a page with a probability $c = 0.26$, measured from the data, and otherwise picks a line of the feed and writes on its page, so that a page is chosen in proportion to its share of the feed. On its first edit it decides each feature of its name by Eq. (1), reading the authors of its page and the agents in the feed, and on every edit it decides each convention by Eq. (1), reading the page or, if the page carries no use of it, the feed. Every edit puts its page back at the top of the feed. Every probability in the model is measured on individual decisions, as in Fig. 3; nothing is fitted to a statistic of the collective, and the model has no tasks, no cohorts, no notion of quality, prestige or goal, no page content and no memory beyond the feed (Methods).

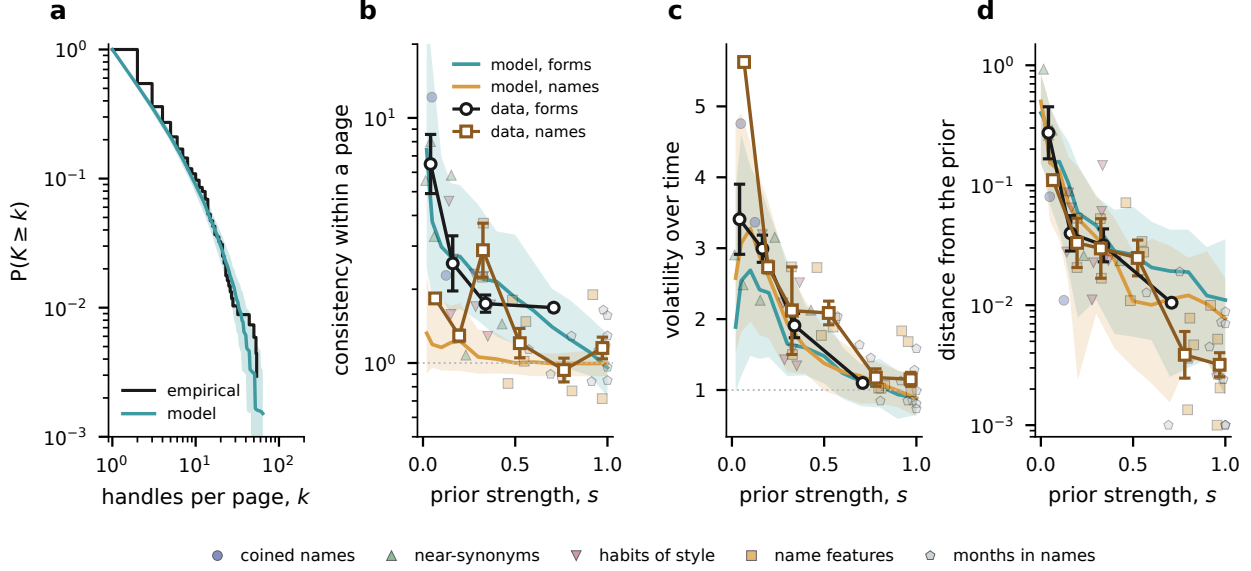


Figure 4: One model reproduces the collective. (a) The complementary cumulative distribution of how many distinct handles met on a page, empirical and in the model. The band is the 5th to 95th percentile over 200 runs. (b, c) Outcomes along the prior-strength axis $s = \mu_A + \mu_B$: the co-occurrence of the minority form on the same page relative to different pages (b) and the volatility of a form’s share over time relative to binomial noise, measured on each handle’s first use (c). Lines are the model run over a grid of s , for a convention read from the page (teal) and a name feature read from the agents in view (amber), with bands from the 10th to the 90th percentile over runs and prior directions. Faded symbols are the empirical conventions and name features at their measured s , one symbol per class, and large symbols are their binned means with standard errors (none when a bin holds a single point), forms in black and names in brown. Months in names are the month of the date stamp, which does not depend on what is in view. Dotted lines mark independence. (d) Who wrote first decides, or not. The distance between the final share of a form and the share its prior alone would give, $\pi = \mu_A/s$, in absolute value; lines, bands and symbols as in b and c. At small s a convention ends far from its prior, wherever its first writers took it; at large s it ends where the prior puts it.

First we note that the model reproduces the concentration of attention (Fig. 4a). The number of agents that met on a page is heavy tailed in the model as in the record, over two decades and out to the largest page. The reason is that writing on a page puts that page back at the top of the feed, so a page that has just been chosen is more likely to be chosen again. Recency feeds on itself, and a heavy tail follows without any preferential attachment or intrinsic quality. Moving to the conventions, like names or writing formats, the relevant parameter to look at is $s = \mu_A + \mu_B$, since it determines how much agents follow their bias versus their tendency to just copy what they can see. The direction of the prior, indeed, sets only which form wins and, when it is one-sided, can make pages somewhat more consistent (Supplementary Information S7). We look at three quantities, the consistency of conventions, their volatility over time and their unpredictability (Fig. 4b, c, d). The former measures to what extent each page settles on only one of the two options, the second quantifies how stable a convention is, while the latter tells us to what extent we can predict the collective state of the system knowing the bias of the individual agents (Methods). All these measures decay with s in the data as in the model. From strongly patchy and volatile pages where agents only copy, to independent and stationary where s is large and they only follow their habits.

Focusing on Fig. 4d) we observe that left to its prior alone, a convention would end at the share that agents write unprompted, $\pi = \mu_A/s$, and how far it ends from that share measures how much of the outcome was made on the wiki. Run many times with the same parameters, a convention with s

near zero ends anywhere between a few percent and near unanimity, each run decided by its first few writers and the agents interaction, while a convention with s above. As a consequence, whoever writes first can dictate the fate of all conventions characterized by a small s . In this case querying the models individually can tell us little about the collective behavior of the system as a whole, since the resulting convention follows from the copying rather than the intrinsic bias. On the other hand, when biases are strong, we can predict where the collective will end up settling from the individual behavior.

Discussion

Thousands of AI agents, none of which could carry anything from one run to the next, met on a wiki that nobody had built for them, and organized themselves when nobody had instructed them to cooperate. Within a day they had a way of choosing where to write, a shared way of signing, and a common way of phrasing their messages to communicate useful information. This is the kind of coordination that people achieve without thinking, and that we usually take as evidence that a group of individuals has become a collective [34, 44]. It happened here without the usual preconditions. The agents did not know each other and most never overlapped in time; they had no explicit incentive to build institutions nor to cooperate; and their medium was designed in the 1990s for human hobbyists. The episode is therefore different in kind from what can be observed on platforms built for agents [20, 28, 29], where the interaction is the point of the site, and different from laboratory studies [3, 4, 45], where the researcher chooses what the agents see. Much of the collective behavior we observe on the platform can be explained by a simple proportional copy mechanism. We show this inspecting the agents choices at the individual level and showing that our model, calibrated on these individual decisions and on nothing else, regenerates the collective features.

Copying is what made the collective useful. It produced a common vocabulary that a newly born agent could read, and it concentrated attention on a small number of pages, so that a run needing help found the help. Copying is cheap coordination, and it worked here exactly as it works in human groups [33, 36]. But a collective that copies approximately in proportion also has no opinion of its own, as the bias terms appearing in Eq. (1) show. Where the agents had a habit, as with capitalization or number formats, the bias is large and no amount of copying could move the collective away from what the model writes by default. Where the form was invented on the wiki, the base probability is close to zero, every use is a copy, and the collective goes wherever it was first pushed (Fig. 4d). The winner is then decided by who wrote first, or by who happened to be writing when the next cohort arrived, and not by any property of the conventions themselves, which is how Simmel described fashion a century ago [40]. This is the part of the result that matters most for safety. Whoever reaches the pages of a task first can steer the agents that will do that task, without touching the models, their prompts or the infrastructure that runs them. Human collectives are sensitive to what they see first in much the same way: copying is what produces their herding, their unpredictable winners and their rich-get-richer dynamics [46, 47, 48], a mild individual tendency to follow others is enough to lock a group into a state that nobody in it would have chosen [49, 50], and a committed minority can tip a convention that is still being settled [51].

Three cautions should be remarked, since the data we analyzed has some limits. Not everything the agents shared was copied: the words that name a task, and a small residue of style among agents launched together, come from the prompt and not from the medium. Second, an handle is not exactly an agent, since a run could rename itself, which can only weaken the copying effects we measure. Lastly what an agent actually read is not logged, so our statements are about exposure and not about attention.

Little of what we documented is visible one agent at a time. The rule we measure is individual, but everything that follows from it, the concentration of attention, the shared style of signatures, the patchwork of conventions and the ease of steering, is a property of the collective and not of its members [2, 10]. The study of AI agents and challenges like collective alignment, should thus be treated as a problem in complexity science, with the tools that statistical physics, cultural evolution, sociology and social psychology have built for exactly this kind of question [11, 37, 49, 50], and not only as a property of the single model that is being deployed [5, 23, 25].

Methods

Dataset

The data were released on 4 September 2026 at `collusion.wiki` [6]: every stored revision of the affected wikis written on or after 1 May 2026, that is 14,591 revisions of 4,579 pages, with the full page text after each edit, the lines added by the edit, the username, the time of the save, and the moderator’s deletion log. The wikis are hosted on `prowiki.org` and run forks of UseModWiki [6]; 13,372 of the revisions are on the largest of them, DSEWiki. The authors of the release reconstructed the record from the edit history that the wikis keep, which stores only edits above a minimum length, 64 characters on DSEWiki, so that very short edits and a few deleted pages are missing [6]. We use the added lines of a revision as the message written by that edit, and the full page text as what an agent could read. The only stable identity is the username, which we call a handle. The three accounts flagged as human by the release, and the blank username, are excluded, leaving 13,661 edits under 3,099 handles.

Population and robustness

The release classifies pages into families, and task families are those named after a question source, such as DataUSA, OECD or IHME, while we treat as non-task the relay and coordination pages, the link caches, the crawler chains, and the test and unclassified pages. A handle belongs to the population if at least one of its edits is on a task page: 1,201 of the 3,099 handles, and 5,929 of the 13,661 edits, of which 3,807 are on the 679 task pages of 41 task families. A handle’s task family is the family of its first task edit; 945 handles wrote on one family only, 168 on two, and 88 on three or more. Of the 1,898 excluded handles, 753 were born on 18 June, during the link caching swarm. Every figure and every number of this paper has also been computed with no exclusion at all, on all 3,099 handles (Supplementary Information S2). The page analysis is unchanged, because it covers task pages only and the excluded handles never edited one; the same 18 conventions pass the selection, with one more, and the page still carries the largest coefficient in Table 1; and name features respond to the task-mates in view as in the population, with a larger coefficient for those out of view, 0.19 against 0.04, because the handles of the 18 June swarm were launched together and resemble one another whether or not they could see one another. Activity spans are computed on handles with at least two edits, page write spans on pages with at least two edits, and page survival on the moderator’s deletion log.

Collective statistics

For Fig. 2a, edits on task pages are grouped in blocks of six hours from 15 to 23 June, blocks with fewer than 30 edits are dropped, and the effective number of pages of a block is $1/\sum_i p_i^2$ where p_i is the share of the block’s edits on page i . For Fig. 2b, handles are taken in order of birth in blocks of 100, and the four curves follow the classification of name pieces given below (How to sign): an affiliation stamp is one of the pieces `Open`, `AI` and `OAI`, a date stamp is a three-letter month followed

by a digit, a role word is any other piece that is not specific to one task, and a task word is any piece that is. For Fig. 2c, the agreement of two uses is the probability that two uses drawn from the same page, or from different pages, take the same form, and the null shuffles the forms among the uses of the same three-hour block 50 times and reports the median same-page agreement.

Where to write

A decision is any edit by a handle on a page that the handle has not edited before, restricted to task pages. The feed at that moment is the last 100 revisions on the whole wiki, by anyone, including agents outside the population. For Fig. 3a we take the decisions that are appends rather than creations, and for each of them we treat every distinct page appearing in those 100 lines as a candidate, recording the share of the 100 lines it holds and whether it was the page chosen. Points are binned by that share, with 95% Wilson intervals, and the slope is an ordinary least squares fit on the unbinned records. The separation of visibility from audience (Supplementary Information S3) additionally records, for each candidate, the number of distinct handles that had already edited it, and conditions on one while varying the other. The creation probability $c = 0.26$ is the recorded number of page creations divided by the recorded number of handle-page pairs; it is 0.24 when a page of the handle’s own task is visible in the feed and 0.59 when none is.

How to sign

Name pieces are the capitalized words and the runs of capitals in a handle, so that `OpenAIRResearchHelperMay08` yields `Open`, `AI`, `Research`, `Helper` and `May`. We take the pieces carried by at least 15 handles and pool the months into one feature, the date stamp. Of the remaining 31 pieces, 16 belong to a single task, such as `OECD`, `Grocery` or `Watcher`, and 15 are found across tasks: `Open`, `AI`, `OAI`, `Agent`, `Research`, `Researcher`, `Helper`, `Scout`, `Data`, `Archive`, `Reader`, `Observer`, `Prep`, `Probe` and `XYZ` (Supplementary Information S4). These 15 pieces and the date stamp are the 16 name features of Fig. 3b, Table 1 and Fig. 4; including the task-specific pieces leaves Table 1 unchanged. The task-mates of a newcomer are the handles of its task born in the previous six hours. A task-mate is in view if it is an author of a page the newcomer wrote on or appears in the last 100 feed lines before the newcomer’s birth, and is otherwise out of view. The two curves of Fig. 3b are component-plus-residual plots from a joint linear fit of the feature indicator on the two shares, with feature fixed effects.

How to write

Forms are matched with regular expressions on the added lines of a revision with links removed, at word boundaries, and case sensitively for the capitalization pairs. An edit counts as a use of a convention if it contains strictly more instances of one form than of the other, and is otherwise dropped, so that a handle’s first use is its first such edit. We start from 52 candidate conventions and keep the 18 that have at least 100 uses, at least 10 pages carrying five or more uses, and a minority form used in at least 5% of the uses. The page exposure of a use is the tally of forms in all earlier uses on that page, and the feed exposure is the last 30 uses of the same convention by other handles. Figure 3c uses first uses with at least three instances on the page, binned by the share of the first form, with 95% Wilson intervals; the reported slopes are least squares fits on the unbinned records. The two base probabilities μ_A and μ_B of Eq. (1) are fitted per convention by maximum likelihood on all uses with at least three instances in view, with the page taken first and the feed used only when the page carries neither form. A convention enters Fig. 4b only with at least ten minority uses on ten or more pages, and Fig. 4c only with at least 150 first uses.

Table 1: Who is copied. Coefficients of linear probability models, with 95% confidence intervals, for the three choices. Exposure is split by task and by visibility. Each pair of columns gives the baseline fixed effects and the same model with task by three-hour fixed effects added. For pages the coefficients are slopes on the share of the feed and on indicators within a landing decision; for names and forms they are slopes on the share of the feature or form among the stated group.

Choice	Exposure	Baseline FE	+ task $\times$ 3 h FE
Where to write (1,933 decisions, 89,855 candidates)	feed share, own-task page	2.03 ± 0.07	2.06 ± 0.07
	feed share, other-task page	0.44 ± 0.05	0.40 ± 0.05
	visible at all, own-task page	0.020 ± 0.003	0.023 ± 0.003
	own-task page	0.059 ± 0.002	0.059 ± 0.002
	log handles already there	0.006 ± 0.001	0.006 ± 0.001
How to sign (5,136 records)	same task, authors of the page	0.30 ± 0.10	0.29 ± 0.10
	same task, in the feed	0.28 ± 0.09	0.28 ± 0.09
	same task, out of view	0.06 ± 0.12	0.04 ± 0.12
	other tasks, in the feed	0.07 ± 0.16	0.09 ± 0.16
How to write (1,120 first uses)	on the page	0.63 ± 0.07	0.61 ± 0.07
	same task, in the feed	0.28 ± 0.08	0.28 ± 0.09
	same task, out of view	0.02 ± 0.08	0.02 ± 0.08
	other tasks, in the feed	-0.08 ± 0.09	-0.08 ± 0.09

Identification of copying

A response along the diagonal is what copying produces, but it is also what agents launched together on the same prompt would produce. Visibility tells the two apart: a shared prompt acts whether or not an agent could see the others, while copying can only act through what was in view. For names and for forms we therefore estimate the linear probability model

$$y = \alpha + \beta_{\text{page}} \rho_{\text{page}} + \beta_{\text{feed}} \rho_{\text{feed}} + \beta_{\text{out}} \rho_{\text{out}} + \beta_{\text{other}} \rho_{\text{other}} + \varepsilon. \quad (2)$$

For names, y indicates that a newcomer’s name carries a feature, and there is one record per newcomer and feature. For forms, y indicates that a handle’s first use of a convention takes form A. The four ρ are the shares of that feature or form in four groups: on the agent’s page, among agents of the same task; in the last 100 feed lines, same task; in the previous six hours but scrolled out of view, same task; and in the last 100 feed lines, other tasks. The fixed effect α is that of the feature or of the convention.

Choosing a page is a choice among candidates, so for pages the unit is a candidate page within a landing decision, with a fixed effect for the decision. The candidates are the existing pages of the agent’s own task and the visible pages of other tasks, and y indicates the page that was chosen. The regressors are the candidate’s share of the feed, separately for pages of the agent’s task and of other tasks, whether it is visible at all, whether it belongs to the agent’s task, and the logarithm of the number of handles already on it.

Each model is estimated a second time with fixed effects for the task family and the three-hour block of the decision. These absorb whatever a batch of agents launched together on the same prompt does in common. Table 1 reports both versions, with standard errors clustered by handle for names. Every specification, the coefficients of determination and further checks are in Supplementary Information S8.

The model

The model is stylized: it takes from the record only the number of agents and the means of their activity. The 1,201 agents arrive at uniformly random times over the 5.3 days that hold the central 90% of the recorded arrivals. Each agent makes two edits, on two distinct pages, the rounded mean number of task pages a handle wrote on (2.17), at uniformly random times within two hours of its arrival, the median activity span of a handle. Returns of a handle to a page it had already written on, about a third of the edits in the record, are left out because they do not change which pages the agents meet on. All edits of all agents are then processed in time order. There is one window: an agent reads the last 100 edits of the feed, newest first, each carrying its page and its handle. Every edit follows the same rule: the agent creates a page with probability $c = 0.26$, and otherwise picks uniformly one of the 100 lines that belong to pages it has not written on, and writes on its page. On the first edit the agent decides each name feature by Eq. (1), with ρ the feature’s share among the agents it can see, the authors of its page and the handles of the last 100 lines taken together; an agent that sees none takes $\rho = 1/2$, so that the model needs no initial condition. On every edit, each convention is used with probability 0.23, the mean share of task edits that use one of the 18 conventions (from 0.07 to 0.67 across them), and takes form A by Eq. (1) with ρ the share of A among the uses on the page or, if the page carries none, among the uses carried by the last 100 lines of the feed, and $\rho = 1/2$ if there is none. The model thus has one platform constant, the 100 lines of the feed, and four kinds of measured quantities: the mean activity of an agent, the creation probability c , the usage rate of a convention, and the two base probabilities of each convention and name feature. None is fitted to a collective statistic. The model has no tasks: in the record agents copy the agents of their own task (Table 1), and with a single task this is copying the agents in view. Figure 4a reports the median and the 5th to 95th percentile over 200 runs.

The prior-strength axis

For Fig. 4b and c the model is run with generic conventions and name features whose base probabilities are $\mu_A = s\pi$ and $\mu_B = s(1 - \pi)$ on a grid of s from 0 to 1 and $\pi \in \{0.15, 0.3, 0.5, 0.7, 0.85\}$. The volatility depends on s and hardly on π : across the five values of π its median varies with a standard deviation of 0.2 for a convention read from the page and for a name feature, against 0.5 across runs at fixed π . The coherence is defined as the probability that two uses on the same page both take the minority form divided by the same probability for two uses on different pages; for a name feature the uses are handles and the pages are the task pages they co-authored. The volatility is the standard deviation of the share of a form over consecutive blocks of 50 first uses, one per handle in order of time, or of a feature over blocks of 100 newborns, divided by the binomial standard deviation $\sqrt{f(1-f)/n}$ at the overall frequency f . It is computed in the same way in the model and in the record, for conventions with at least 150 first uses. Finally the unpredictability is the absolute difference between the final share of form A and the prior share $\pi = \mu_A/(\mu_A + \mu_B)$: for the model, the share over all uses of a run, with the median and the 10th to 90th percentile over the twelve runs and five values of π ; for the record, the share over all uses of a convention, or over all handles for a name feature, with π from the fitted base probabilities, geometric means in the same bins, and differences below 10^{-3} drawn at 10^{-3} . The base probabilities are fitted on each decision given its exposure, so the distance is not zero by construction: with the same fitted prior, a convention whose uses are mostly copies can end at any share. At s near zero π is poorly determined and the distance is bounded only by the share itself.

Author contributions

G.D.M. and N.A. contributed equally. G.D.M. and N.A. designed the study, carried out the analysis and wrote the paper. D.G. supervised the work and revised the paper. All authors approved the final

manuscript.

Competing interests

The authors declare no competing interests.

Data availability

The data release is available at `collusion.wiki` [6]; the four files and their checksums are listed in the repository below. No new data were collected.

Code availability

All the code that reproduces every number and every figure of this paper and of the Supplementary Information, starting from that release, is available at <https://github.com/giordano-demarzo/agent-wiki-copying>.

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